

Windows Event Log Investigation

Overview

This assignment focused on analyzing a simulated Windows Event Log dataset using Splunk to identify system patterns, potential issues, and gain insights into typical Windows operations. The dataset contained 100 XML-formatted entries representing various types of Windows system, application, and security events.

The primary goal of the assignment was to develop hands-on skills in ingesting Windows logs, extracting relevant fields, performing searches and statistical analysis, visualizing data, and identifying anomalies or security-related behaviors.

Data Ingestion and Field Extraction

The dataset (`windows_event_log_example_fixed.xml`) was successfully uploaded into Splunk and indexed under `windows_log`. The log entries were structured in XML, which required the use of Splunk's `spath` command to extract nested fields such as:

- `EventID` – the unique identifier for each Windows event
- `Level` – the severity level (e.g., Information, Warning, Error)
- `SourceName` – the system or application generating the log
- `TaskCategory` – the context or action related to the event
- `Message` – a brief description of the event

Splunk did not automatically extract all necessary fields due to the XML format, so custom field extraction using `spath` was performed for accurate analysis.

Search and Analysis

Several SPL queries were used to understand and explore the data:

- **Event count by level** showed the distribution of severity:
 - Information events dominated the logs.

- Warning and Error events were fewer but more critical.
- **Top sources and event IDs** were identified using top and stats commands, revealing frequent contributors like Service Control Manager, Security-Auditing, and Windows Update Client.
- **Time-based trends** were visualized using timechart, which highlighted when bursts of certain event types occurred, such as clusters of warnings or errors in close time frames.

Event Correlation

Event correlation was attempted using the **transaction** command to find sequences of events that may be related. Patterns were identified where certain warning events were followed by errors, suggesting possible causes or escalating issues.

The screenshot shows the Splunk Search interface. The search bar contains the query `index=*windows_log*`. Below the search bar, it indicates 101 events were found. The results are displayed in a table with columns for Time and Event. The first event is from 10/04/2025 at 18:34:36.000, and the second is from 25/02/2024 at 17:50:08.000. The interface includes a search bar, navigation tabs, and a list of fields on the left.

i	Time	Event
>	10/04/2025 18:34:36.000	<?xml version="1.0" ?> <Events> <Event> host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>Application</Source> <EventID>1886</EventID> <TaskCategory>Configuration</TaskCategory> <Level>Error</Level> Show all 8 lines host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>System</Source> <EventID>1583</EventID> <TaskCategory>Startup/Shutdown</TaskCategory> <Level>Error</Level> Show all 8 lines host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation

localhost:8000/en-GB/app/search/search?q=search%3Dindex%3D%22windows_log%22&earliest=0&latest=&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=&sid=1744335343.279#

Splunk Analysis Assignment: xSearch | Splunk 9.3.1x+

localhost:8000/en-GB/app/search/search?q=search%20index%3Dwindows_log%20%7C%20timechart%20span%3D1h%20count%20by%20Level&earliest=0&latest=&di...

GmailYouTubeNew chatimage.pnghttps://portal.ustra...Credlia's Student...my.csudh.edu CS...MapsCertified AppSec...Input Validation M...All Bookmarks

splunk>enterpriseAppsMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

New SearchSave AsCreate Table ViewClose

1index=windows_log | timechart span=1h count by LevelAll time

✓ 101 events (before 10/04/2025 18:40:00.000)No Event SamplingJobSmart Mode

EventsPatternsStatistics (9,841)Visualization

20 Per PageFormatPreview< Prev12345678...Next>

_time	count
2024-02-25 17:00	100
2024-02-25 18:00	0
2024-02-25 19:00	0
2024-02-25 20:00	0
2024-02-25 21:00	0
2024-02-25 22:00	0
2024-02-25 23:00	0
2024-02-26 00:00	0
2024-02-26 01:00	0
2024-02-26 02:00	0
2024-02-26 03:00	0
2024-02-26 04:00	0
2024-02-26 05:00	0
2024-02-26 06:00	0
2024-02-26 07:00	0

Splunk Analysis Assignment: xSearch | Splunk 9.3.1x+

localhost:8000/en-GB/app/search/search?q=search%20index%3Dwindows_log%20%7C%20spath%20input%3D_raw%20%7C%20stats%20count%20by%20Level&earli...

GmailYouTubeNew chatimage.pnghttps://portal.ustra...Credlia's Student...my.csudh.edu CS...MapsCertified AppSec...Input Validation M...All Bookmarks

splunk>enterpriseAppsMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

New SearchSave AsCreate Table ViewClose

1index=windows_log | spath input=_raw | stats count by LevelAll time

✓ 101 events (before 10/04/2025 18:42:21.000)No Event SamplingJobSmart Mode

EventsPatternsStatistics (3)Visualization

20 Per PageFormatPreview

Level	count
Error	42
Information	24
Warning	34

Splunk Analysis Assignment: xSearch | Splunk 9.3.1x+

localhost:8000/en-GB/app/search/search?q=search%20index%3Dwindows_log%20%7C%20table%20_time%20_raw&earliest=0&latest=&display.page.search.mode=sm...All Bookmarks

splunk>enterpriseApps

MessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

New Search

Save AsCreate Table ViewClose

1index=windows_log | table _time _rawAll time

101 events (before 10/04/2025 18:42:42.000)No Event SamplingJobSmart Mode

EventsPatternsStatistics (101)Visualization

20 Per PageFormatPreview

_time_raw

2024-02-25 17:58:08<DateTime>2024-02-25 17:58:08</DateTime><Source>System</Source><EventID>1868</EventID><TaskCategory>Configuration</TaskCategory><Level>Error</Level><Description>Security permissions were updated.</Description></Event><Event><DateTime>2024-02-25 17:58:08</DateTime><Source>Application</Source><EventID>1485</EventID><TaskCategory>Startup/Shutdown</TaskCategory><Level>Information</Level><Description>Security permissions were updated.</Description></Event><Event><DateTime>2024-02-25 17:58:08</DateTime><Source>Security</Source><EventID>1071</EventID><TaskCategory>Startup/Shutdown</TaskCategory><Level>Error</Level><Description>Application error occurred.</Description></Event>

Visualizations and Dashboarding

Splunk Analysis Assignment: xwindows_log_analysis | Splunk xwindows_log_analysis | Splunk xwindows_log_analysis Copy x+

localhost:8000/en-GB/app/search/windows_log_analysisAll Bookmarks

splunk>enterpriseApps

MessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

windows_log_analysis

EditExport...

NULL_span

2024-02-25T17:00:00.000-08:00100 (2.7%)

2024-02-25T18:00:00.000-08:003600 (100%)

extracted xml file

i	Time	Event
>	10/04/2025 18:34:36.000	<?xml version="1.0" ?><Events><Event>host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateTime>2024-02-25 17:58:08</DateTime><Source>Application</Source><EventID>1886</EventID><TaskCategory>Configuration</TaskCategory><Level>Error</Level>

Show all 8 lines

Splunk Analysis Assignment: x windows_log_analysis | Splunk x windows_log_analysis | Splunk x windows_log_analysis Copy x +

localhost:8000/en-GB/app/search/windows_log_analysis

extracted xml file

i	Time	Event
>	10/04/2025 18:34:36.000	<?xml version="1.0" ?> <Events> <Event> host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>Application</Source> <EventID>1886</EventID> <TaskCategory>Configuration</TaskCategory> <Level>Error</Level> Show all 8 lines host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>System</Source> <EventID>1503</EventID> <TaskCategory>Startup/Shutdown</TaskCategory> <Level>Error</Level> Show all 8 lines host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>System</Source> <EventID>1888</EventID> <TaskCategory>Startup/Shutdown</TaskCategory> <Level>Warning</Level> Show all 8 lines host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>Security</Source> <EventID>1782</EventID> <TaskCategory>Configuration</TaskCategory> <Level>Error</Level> Show all 8 lines host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation

localhost:8000/en-GB/app/search/windows_log_analysis#?time=2024-02-25%2017:50:08%2FDateandTime%3E

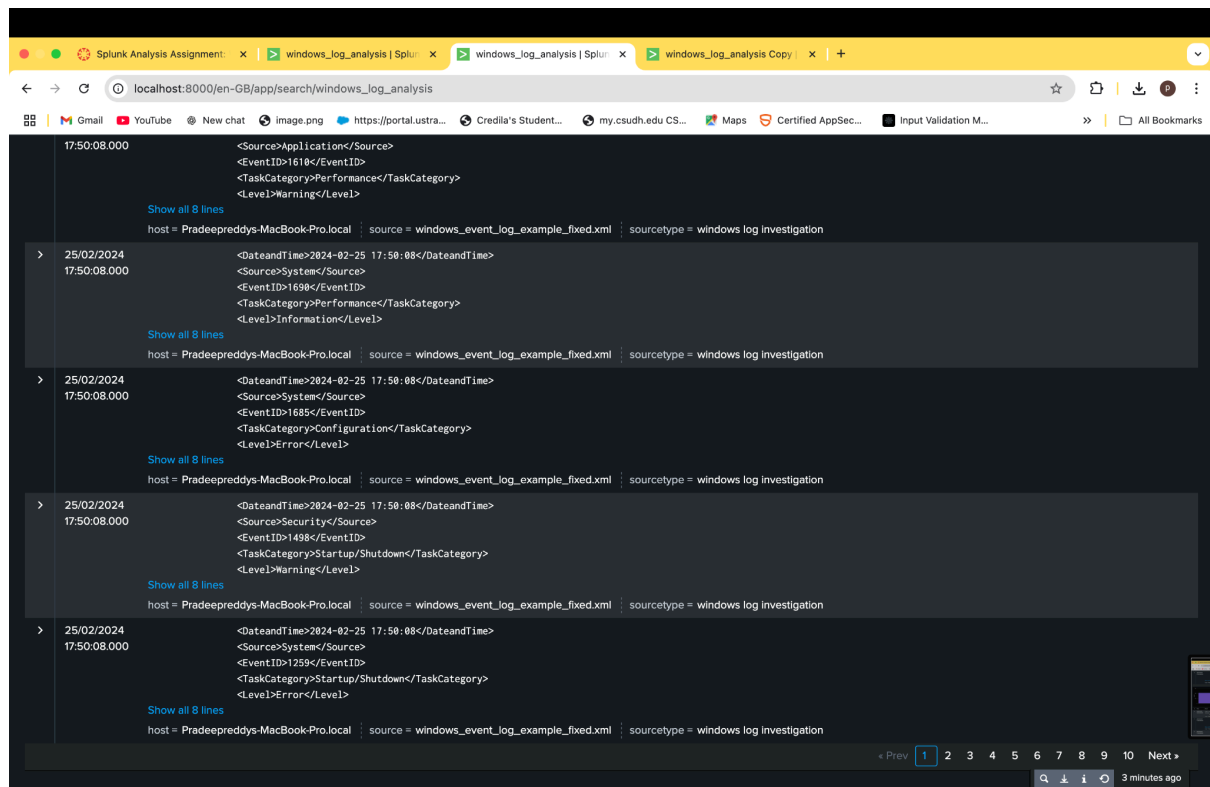
Splunk Analysis Assignment: x windows_log_analysis | Splunk x windows_log_analysis | Splunk x windows_log_analysis Copy x +

localhost:8000/en-GB/app/search/windows_log_analysis

>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>System</Source> <EventID>1259</EventID> <TaskCategory>Startup/Shutdown</TaskCategory> <Level>Error</Level> Show all 8 lines host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
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< Prev 1 2 3 4 5 6 7 8 9 10 Next >

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>	10/04/2025 18:34:36.000	<?xml version="1.0" ?> <Events> <Event> host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>Application</Source> <EventID>1886</EventID> <TaskCategory>Configuration</TaskCategory> <Level>Error</Level> Show all 8 lines host = Pradeepreddys-MacBook-Pro.local source = windows_event_log_example_fixed.xml sourcetype = windows log investigation
>	25/02/2024 17:50:08.000	<DateandTime>2024-02-25 17:50:08</DateandTime> <Source>System</Source> <EventID>1503</EventID>



Reflection

A key challenge in this assignment was dealing with XML log formatting, which required manual field extraction using `spath`. Understanding the structure of Windows Event Logs and leveraging Splunk's capabilities for parsing and analysis was critical. Overcoming these technical hurdles greatly enhanced confidence in working with real-world log data and building automated monitoring systems.